

VISTRA ENERGY



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CP-201900192
TXX-19040

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Ref 10 CFR 50.73

04/25/2019

SUBJECT: COMANCHE PEAK NUCLEAR POWER PLANT
DOCKET NOS. 50-445 & 50-446
LTOP POWER OPERATED RELIEF VALVE (PORV) SETPOINT
LICENSEE EVENT REPORT 445/19-001-00

Dear Sir or Madam:

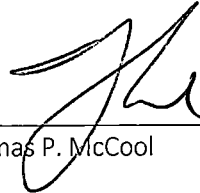
Pursuant to 10CFR50.73, Vistra Operations Company LLC (Vistra OpCo), hereby submits the enclosed Licensee Event Report 445/19-001-00, "LTOP Power Operated Relief Valve (PORV) Setpoint" for Comanche Peak Nuclear Power Plant (CPNPP), Units 1 and 2.

This communication contains no new licensing basis commitments regarding CPNPP Units 1 and 2.

IEZZ
NRR

If you have any questions regarding this submittal, please contact Garry Struble at (254) 897-6628 or garry.struble@luminant.com.

Sincerely,



Thomas P. McCool

Enclosure COMANCHE PEAK NUCLEAR POWER PLANT
 LTOP POWER OPERATED RELIEF VALVE (PORV) SETPOINT
 LICENSEE EVENT REPORT 445/19-001-00

c – Scott Morris, Region IV
 Natreon Jordan, NRR
 Resident Inspectors, Comanche Peak

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Comanche Peak Nuclear Power Plant	2. Docket Number 05000 445	3. Page 1 OF 3
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4. Title LTOP Power Operated Relief Valve (PORV) Setpoint

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	13	2019	2019	001	00	04	25	2019	CPNPP Unit 2	05000 446
									Facility Name	Docket Number
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact Garry Struble	Telephone Number (Include Area Code) 254-897-6628
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date)					☒ No					

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On February 2, 2019 at 1500, Westinghouse reported that while reviewing the Low Temperature Overpressure (LTOP) system analysis of record (CN-SCS-04-100, Rev. 3) for areas of margin that could offset the increase in differential pressure required to support the Mechanical Stress Improvement Project (MSIP), errors were discovered. On March 9, 2019 an evaluation was completed that determined the Unit 2 LTOP system had been inoperable in the past for longer than allowed by Technical Specifications. Corrective actions include changing the LTOP PORV lift settings, on both Units 1 and 2 to meet the protection requirements of 10 CFR 50 Appendix G, Fracture Toughness Requirements.

All times listed in this report are Central Standard Time (CST).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	YEAR	3. LER NUMBER	REV NO.
Comanche Peak Nuclear Power Plant	05000- 445	2019	- 001	- 00

NARRATIVE**I. DESCRIPTION OF REPORTABLE EVENT****A. REPORTABLE EVENT CLASSIFICATION**

This event is reportable in accordance with 10CFR50.73(a)(2)(i)(B);

Any operation or condition which was prohibited by the plant's Technical Specifications except when:

- (1) The Technical Specification is administrative in nature;
- (2) The event consisted solely of a case of a late surveillance test where the oversight was corrected, the test was performed, and the equipment was found to be capable of performing its specified safety functions; or
- (3) The Technical Specification was revised prior to discovery of the event such that the operation or condition was no longer prohibited at the time of discovery of the event.

B. PLANT CONDITION PRIOR TO THE EVENT

Both Unit 1 and Unit 2 were operating at 100% Rated Thermal Power.

C. STATUS OF STRUCTURES, SYSTEMS, OR COMPONENTS THAT WERE INOPERABLE AT THE START OF THE EVENT AND CONTRIBUTED TO THE EVENT

Prior to the event, no Structures, Systems, or Components were inoperable that contributed to the event.

D. NARRATIVE SUMMARY OF THE EVENT, INCLUDING DATES AND APPROXIMATE TIMES

On February 4, 2019, Westinghouse informed Comanche Peak that calculation errors had caused the maximum allowable PORV LTOP lift settings to be higher than allowed to meet 10 CRR 50 Appendix G, Fracture Toughness Requirements.

On March 13, 2019, an evaluation was completed that determined the Unit 2 LTOP system was inoperable for greater than the 7 day Completion Time while in MODE 4 on one occasion. The LTOP system was not operable because it did not comply with 10 CFR 50, Appendix G, Fracture Toughness Requirements. This single event would have required depressurizing the Reactor Coolant System (RCS) and establishing an RCS vent ≥ 2.98 square inches within 8 hours. The evaluation also determined the Unit 2 LTOP system was inoperable for greater than the 24 hour Completion Time on three occasions while in MODE 5. The LTOP system was not operable because it did not comply with 10 CFR 50, Appendix G, Fracture Toughness Requirements. Each of the three events would have required depressurizing the Reactor Coolant System (RCS) and establishing an RCS vent ≥ 2.98 square inches within 8 hours.

In all four cases the failure to vent the RCS caused operation in a manner prohibited by Technical Specifications. The current PORV LTOP lift settings would not have resulted in a challenge to reactor vessel integrity and would have performed their safety function

E. THE METHOD OF DISCOVERY OF EACH COMPONENT OR SYSTEM FAILURE, OR PROCEDURAL PERSONNEL ERROR

A past operability evaluation was completed on March 13, 2019 that identified four times that Unit 2 was operating outside of Technical Specification requirements associated with completion times for LTOP inoperability in the three years prior to event discovery. Unit 1 was not operated outside of Technical Specifications in the three years prior to event discovery.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Comanche Peak Nuclear Power Plant	05000- 445	YEAR 2019	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE

II. COMPONENT OR SYSTEM FAILURES

There were no component or system failures.

A. CAUSE OF EACH COMPONENT OR SYSTEM FAILURE

Not applicable.

B. FAILURE MODE, MECHANISM, AND EFFECTS OF EACH FAILED COMPONENT

Not applicable.

C. SYSTEMS OR SECONDARY FUNCTIONS THAT WERE AFFECTED BY FAILURE OF COMPONENTS WITH MULTIPLE FUNCTIONS

Not applicable.

D. FAILED COMPONENT INFORMATION

There were no failed components.

III. ANALYSIS OF THE EVENT

A. SAFETY SYSTEM RESPONSES THAT OCCURRED

No safety system responses occurred due to this event.

B. DURATION OF SAFETY SYSTEM TRAIN INOPERABILITY

- Unit 1 had no LTOP inoperabilities due to the event.
- Unit 2 had one inoperability that exceeded 7 days while in MODE 4 on 6/13/2017.
- Unit 2 had three separate inoperabilities that exceeded 24 hours while in MODE 5 on 4/29/17, 7/29/17, and 1/9/19.

C. SAFETY CONSEQUENCES AND IMPLICATIONS OF THE EVENT

Based on Westinghouse and Comanche Peak evaluation LTOP was inoperable four different times as described earlier in this report. Those evaluations also indicate that LTOP was capable of protecting the reactor vessel from low temperature overpressure failure even with the less restrictive LTOP PORV lift settings as described in this report. Therefore, this event had no impact on nuclear safety, reactor safety, radiological safety, or health and safety of the public. This event has been evaluated to not meet the definition of a safety system functional failure per 10 CFR 50.73(a)(2)(v).

IV. CAUSE OF THE EVENT

Incorrect PORV LTOP lift settings were calculated due to inappropriate application of uncertainties causing incorrect lift settings to be placed in the PTLR. Those incorrect settings lead to non-compliance with 10 CFR 50 Appendix G, Fracture Toughness Requirements which caused the four LTOP inoperabilities.

V. CORRECTIVE ACTIONS

Recalculate PORV LTOP lift settings utilizing the correct uncertainties, update the PTLR, and adjust plant installed PORV LTOP lift settings in accordance with the revised PTLR.

IV. PREVIOUS SIMILAR EVENTS

Not Applicable